

Adri Katyayan

+1-(551)312-9658 | adri.katyayan@stonybrook.edu | linkedin.com/in/adri-katyayan-21a0b2222/ | github.com/ad-kat

EDUCATION

Stony Brook University

Master's in Computer Science – CGPA: 3.44

Stony Brook, NY

Aug. 2025 – May 2027

Manipal Institute of Technology

B.Tech in Computer and Communication Engineering, Minor in Computational Intelligence

Udupi, India

Oct. 2021 – Jul. 2025

TECHNICAL SKILLS

Languages: C++, Python, Java, Rust, SQL, JavaScript/TypeScript

AI/ML: PyTorch, TensorFlow (Keras), HuggingFace Transformers, Scikit-learn, LLM APIs (Groq/Llama), Prompt Engineering, Knowledge Distillation, NLP, CNN

Backend & Cloud: FastAPI, Flask, React, REST APIs, PostgreSQL, MySQL, SQLite, Docker, Vercel, Git, Kubernetes (basic), AWS (EC2/S3/RDS), GCP (BigQuery/Compute Engine), GitHub Actions (CI/CD)

Systems & Networking: Multithreading, WebSockets, Distributed Systems, Performance Optimization

WORK EXPERIENCE

DevRev

Jun. 2026 – Present

Member of Technical Staff (MTS) Intern

Palo Alto, CA

- Engineered a latency-aware thinking-status state machine for Kilmer's AI agent with a 600ms linger window, eliminating UI flicker across multi-second turns.
- Integrated Deepgram's Nova-2 streaming API for real-time speech-to-text, chunking audio every 250ms; now extending it into a full voice-agent mode with text-to-speech replies.
- Shipped an end-to-end bug-reporting pipeline with two-step artifact uploads and automated DevRev ticket creation; now migrating 17 NeonDB-backed routes onto DevRev's internal platform.

Stony Brook University

Jan. 2026 – May 2026

Graduate Research Assistant

Stony Brook, NY

- Built a multimodal pipeline fusing Whisper transcription, YAMNet audio classification, and MediaPipe pose tracking, computing 8 body movement metrics per second per performance.
- Designed an LLM punchline detector (Groq Llama-3.3-70B) with disk caching to cut redundant calls, and shipped a FastAPI backend serving 13 endpoints to a live analytics dashboard.

PROJECTS

C++ Limit Order Book & Matching Engine | *C++20, FastAPI, WebSockets, Docker*

Dec. 2025 – May 2026

- Engineered a thread-safe C++20 matching engine (`std::shared_mutex`) with price-time priority FIFO logic, sustaining 1.9M operations per second at 0.4 microsecond latency.
- Cut order cancellation from $O(n)$ to $O(1)$ using hash-indexed list iterators, replacing a naive linear scan across resting orders.
- Built a FastAPI WebSocket bridge streaming 5 live NASDAQ tickers into the engine, paired with a Groq Llama powered commentary agent and React/Vite dashboard, deployed on Hugging Face Spaces.

ClinicalContradiction | *Python, DistilBERT, Groq/Llama, FastAPI, PostgreSQL, spaCy, Docker*

Mar. 2026 – May 2026

- Built a clinical NLP system flagging cross-encounter contradictions across 145,914 MIMIC-IV patients and 331,793 discharge notes under PhysioNet credentialed access.
- Combined a rule-based spaCy extractor with a Groq Llama-3.3-70B layer for higher recall on free-text notes, then distilled both into a DistilBERT student.
- Achieved F1 0.731 binary and F1 0.629 type classification at 7ms inference, 140x faster than the LLM teacher; shipped 8 endpoints via Docker.

Cloud Resource Lifecycle Manager | *Python, FastAPI, PostgreSQL, Docker, APScheduler*

Mar. 2026 – May 2026

- Built an 18-endpoint FastAPI governance API modeling cloud resource lifecycles, auto-stopping TTL breaches and flagging stuck provisioning past 10 minutes every 5 minutes (live on Render).
- Added CSV export for cost reports and full audit trails, plus a 14-test pytest suite wired to GitHub Actions CI on every push.
- Implemented per-cost-centre budget alerting with configurable thresholds (default 80%) that forecasts monthly overspend and flags critical breaches before they occur.